MTH 200 Strategies for Computing Limits

Directions: The goal of the following problems is to help challenge and deepen your understanding of the current topic. Don't assume that you will know how to answer each and every question. You will work with at least two other students in the class to try and find solutions and write arguments for each of the following questions. Be sure that you question your classmates' reasoning. If you don't understand how they are getting something you should ASK THEM TO EXPLAIN! Being able to defend your reasoning is important and so you are actually helping them by asking them to explain.

Challenge Questions

During the past several days we have seen several different techniques that have helped us to calculate limits. Some of the things we have use are:

- Using Limit Laws
- Direction substitution with polynomials and rational functions.
- Rewriting the given expression
  - Factoring
  - Multiplying out quantities
  - Reducing fractions
  - Multiplying by the Conjugate
  - Checking the limit from both sides to see if a limit DNE.
  - Use Trig. identities.
- Using Squeeze Theorem

Consider using these various techniques to help answer each of the following limits.

1. Compute the limit $\lim_{h \to 0} \frac{(2+h)^2 - 4}{h}$

$$\lim_{h \to 0} \frac{(2+h)^2 - 4}{h} = \lim_{h \to 0} \frac{4 + 4h + h^2 - 4}{h} = \lim_{h \to 0} (4 + h) = 4$$

2. $\lim_{u \to 2} \frac{\sqrt{u+7} - 3}{u - 2}$

$$\lim_{u \to 2} \frac{\sqrt{u+7} - 3}{u - 2} = \lim_{u \to 2} \frac{\sqrt{u+7} - 3}{u - 2} \cdot \frac{\sqrt{u+7} + 3}{\sqrt{u+7} + 3} = \lim_{u \to 2} \frac{u + 7 - 9}{(u-2)\left(\sqrt{u+7} + 3\right)} = \lim_{u \to 2} \frac{1}{\sqrt{u+7} + 3} = \frac{1}{6}$$

3. $\lim\limits_{x\to -3^-} \dfrac{81-x^4}{x+3}$

$$\lim_{x\to -3^-} \frac{81-x^4}{x+3} = \lim_{x\to -3^-} \frac{\left(9-x^2\right)\left(9+x^2\right)}{x+3} = \lim_{x\to -3^-} \frac{\left(3-x\right)\left(3+x\right)\left(9+x^2\right)}{x+3} = \lim_{x\to -3^-} \left(3-x\right)\left(9+x^2\right)$$

$$= \left(3-(-3)\right)\left(9+(-3)^2\right) = (6)(18)$$

$$= 108$$

Follow-up: Does it matter that this is a limit from the left? Should this affect the answer?
Answer: No, if the limit had just approached 3 we would have done the same work and then since the limit would have equaled 108 it must then also equal 108 from both sides.

4. $\lim\limits_{x\to -1} \dfrac{x^2}{x^3-x}$

$$\lim_{x\to -1} \frac{x^2}{x^3-x} = \lim_{x\to -1} \frac{x}{x^2-1} = \lim_{x\to -1} \frac{x}{(x-1)(x+1)}$$

We cannot directly substitute here, and there is nothing left to simplify, so we will check the limit from both sides.

$$\lim_{x\to -1^-} \frac{x \;\to -1}{x^2-1 \to 0^+} \to -\infty \quad \& \quad \lim_{x\to -1^+} \frac{x \;\to -1}{x^2-1 \to 0^-} \to \infty$$

Therefore, the limit DNE.

5. $\lim\limits_{t\to 0} \left(\dfrac{1}{t^3} - \dfrac{t-1}{t^4}\right)$

$$\lim_{t\to 0} \left(\frac{1}{t^3} - \frac{t-1}{t^4}\right) = \lim_{t\to 0} \left(\frac{t}{t^4} - \frac{t-1}{t^4}\right) = \lim_{t\to 0} \left(\frac{t-t+1}{t^4}\right) = \lim_{t\to 0} \left(\frac{1}{t^4}\right)$$

We cannot directly substitute here, and there is nothing left to simplify, so we will check the limit from both sides.

$$\lim_{t\to 0^-} \left(\frac{1}{t^4}\right) \frac{\to 1}{\to 0^+} \to \infty \quad \& \quad \lim_{t\to 0^+} \left(\frac{1}{t^4}\right) \frac{\to 1}{\to 0^+} \to \infty$$

Therefore, the limit DNE but we say that $\lim\limits_{t\to 0} \left(\dfrac{1}{t^3} - \dfrac{t-1}{t^4}\right) = \infty$

6. If $\dfrac{x^2-5x+6}{x-2} \le g(x) \le x^2-3x+1$ for all x, determine $\lim\limits_{x\to 2} g(x)$

We see that $\lim\limits_{x\to 2}\dfrac{x^2-5x+6}{x-2} = \lim\limits_{x\to 2}\dfrac{(x-2)(x-3)}{x-2} = \lim\limits_{x\to 2}(x-3) = -1$ and $\lim\limits_{x\to 2}(x^2-3x+1) = (2)^2-3(2)+1 = -1$

Therefore, by the Squeeze Theorem we know that $\lim\limits_{x\to 2} g(x) = -1$.

7. $\lim\limits_{\theta\to 0}\dfrac{\sin(\theta)}{\sec(\theta)\sin(2\theta)}$

$$\lim_{\theta\to 0}\frac{\sin(\theta)}{\sec(\theta)\sin(2\theta)} = \lim_{\theta\to 0}\frac{\sin(\theta)\cos(\theta)}{2\sin(\theta)\cos(\theta)} = \lim_{\theta\to 0}\frac{1}{2} = \frac{1}{2}$$

NOTE! Direction substitution into trig functions hasn't been established yet, so we need to make sure that we don't attempt to directly substitute the value of 0 here!

8. $\lim\limits_{x\to 1}(x-1)^4\cos^2\left(\dfrac{3+x}{x^2-1}\right)$

We note that $-1 \le \cos\left(\dfrac{3+x}{x^2-1}\right) \le 1$.

This implies that $0 \le \cos^2\left(\dfrac{3+x}{x^2-1}\right) \le 1$.

This implies that $0 \le (x-1)^4\cos^2\left(\dfrac{3+x}{x^2-1}\right) \le (x-1)^4$

We see that $\lim\limits_{x\to 1} 0 = 0$ & $\lim\limits_{x\to 1}(x-1)^4 = 0$

Therefore, by the Squeeze Theorem, we know that $\lim\limits_{x\to 1}(x-1)^4\cos^2\left(\dfrac{3+x}{x^2-1}\right) = 0$

9. Consider this piecewise function $f(x)=\begin{cases}\dfrac{\sqrt{x+1}-1}{x}, & x<0\\ \dfrac{x}{x-2}, & 0\le x<2\\ \dfrac{x-2}{x^3-2x^2+x-2}, & x\ge 2\end{cases}$.

NOTE: The answers below are just one possible way these questions could be answered. There may exist more possible solutions.

a) Create a limit for $f(x)$ that equals $\dfrac{1}{2}$.

$$\lim_{x\to 0^-} f(x)=\lim_{x\to 0^-}\frac{\sqrt{x+1}-1}{x}=\lim_{x\to 0^-}\frac{\sqrt{x+1}-1}{x}\cdot\frac{\sqrt{x+1}+1}{\sqrt{x+1}+1}=\lim_{x\to 0^-}\frac{(x+1)-1}{x\left[\sqrt{x+1}+1\right]}=\lim_{x\to 0^-}\frac{1}{\sqrt{x+1}+1}=\frac{1}{2}$$

$$\lim_{x\to 0^+} f(x)=\lim_{x\to 0^+}\frac{x}{x-2}=\frac{0}{-2}=0$$

$$\lim_{x\to 2^+} f(x)=\lim_{x\to 2^+}\frac{x-2}{x^3-2x^2+x-2}=\lim_{x\to 2^+}\frac{x-2}{x^2(x-2)+(x-2)}=\lim_{x\to 2^+}\frac{x-2}{(x^2+1)(x-2)}=\lim_{x\to 2^+}\frac{1}{x^2+1}=\frac{1}{5}$$

b) Create a limit for $f(x)$ that equals -1.

$$\lim_{x\to 1} f(x)=\lim_{x\to 1}\frac{x}{x-2}=\frac{1}{-1}=-1$$

c) Create a limit for $f(x)$ that equals $\dfrac{1}{4}$.

$$\lim_{x\to 8} f(x)=\lim_{x\to 8}\frac{\sqrt{x+1}-1}{x}=\frac{\sqrt{9}-1}{8}=\frac{2}{8}=\frac{1}{4}$$

d) Create a limit for $f(x)$ that DNE.

$$\lim_{x\to 0^-} f(x)=\lim_{x\to 0^-}\frac{\sqrt{x+1}-1}{x}=\lim_{x\to 0^-}\frac{\sqrt{x+1}-1}{x}\cdot\frac{\sqrt{x+1}+1}{\sqrt{x+1}+1}=\lim_{x\to 0^-}\frac{(x+1)-1}{x\left[\sqrt{x+1}+1\right]}=\lim_{x\to 0^-}\frac{1}{\sqrt{x+1}+1}=\frac{1}{2}$$

$$\lim_{x\to 0^+} f(x)=\lim_{x\to 0^+}\frac{x}{x-2}=\frac{0}{-2}=0$$

Therefore, $\lim_{x\to 0} f(x)$ DNE.

e) Create a limit for $f(x)$ that equals negative infinity.

$$\lim_{x\to 2^-} f(x)=\lim_{x\to 2^-}\frac{x}{x-2}\begin{matrix}\to\\ \to\end{matrix}\frac{2}{0^-}\to -\infty\text{, therefore } =\lim_{x\to 2^-} f(x)=-\infty$$

f) Create a limit for $f(x)$ that equals $\frac{1}{101}$.

$$\lim_{x\to 10} f(x)=\lim_{x\to 10}\frac{x-2}{x^3-2x^2+x-2}=\lim_{x\to 10}\frac{x-2}{x^2(x-2)+(x-2)}=\lim_{x\to 2^+}\frac{x-2}{(x^2+1)(x-2)}=\lim_{x\to 10}\frac{1}{x^2+1}=\frac{1}{101}.$$